IE 306 — Systems Simulation

Assignment 2: Coordinated Traffic Corridor Simulation

(Group Assignment — Teams of 3)

Spring 2026 | Due Date: April 7th, 2026

Overview

In this assignment you will build a discrete-event simulation of a **two-intersection traffic corridor** using SimPy. The corridor runs **south-to-north** (S→N); one-way cross-streets run **west-to-east** (W→E) at each intersection. Your model must faithfully represent signal phasing, turning movements, downstream blocking, transit signal priority for buses, and emergency-vehicle preemption. You will then design and run experiments comparing *uncoordinated* vs. *coordinated* signal control and analyze the impact of emergency-vehicle preemptions.

Choosing the right SimPy constructs for each aspect of the system is part of the assignment. Your report should justify your design decisions.

1 System Description

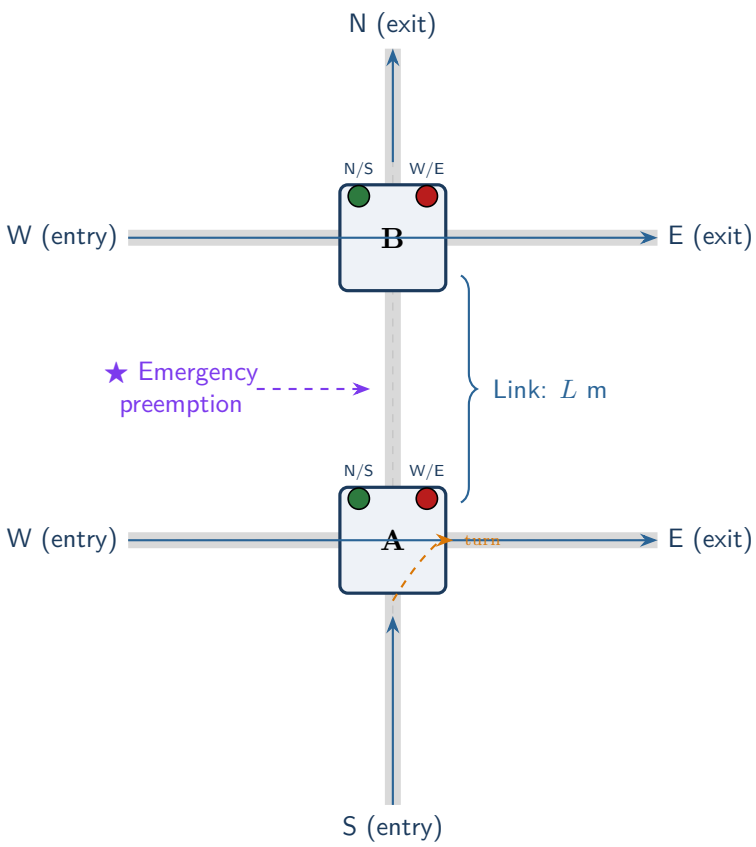

Figure 1: Two-intersection corridor layout. Main flow runs S→N; cross-streets run W→E. Each intersection has two signal phases (N/S and W/E). Turning vehicles exit eastward.

1.1 Corridor Geometry

The corridor consists of two signalised intersections, **A** (upstream/south) and **B** (downstream/north), connected by a single directional link of length $L = 300\,\text{m}$ (see Figure 1). Vehicles enter the system from the south at Intersection A and, if travelling through, proceed northward to Intersection B and exit north. Cross-street (W→E) traffic arrives from the west independently at each intersection.

1.2 Signal Control

Each intersection operates a **two-phase fixed-time signal cycle**:

Table 1: Signal timing parameters.

Parameter	**Intersection A**	**Intersection B**
Cycle length, C	90 s	90 s
N/S green, g_{NS}	45 s	45 s
Amber / all-red clearance	4 s	4 s
W/E green, g_{WE}	37 s	37 s

The phases cycle as: N/S green → amber → W/E green → amber → repeat. The **amber interval** (4 s) is an *all-red clearance* phase: it represents the real-world yellow-plus-all-red period during which the previous phase's traffic clears the intersection before the next phase begins. During this interval **no vehicles may enter** the intersection from *any* direction. A vehicle that is already crossing (i.e. currently being served) finishes normally, but no new service may start.

Coordination. In the *coordinated* scenario, Intersection B's N/S green phase starts with an **offset** ϕ relative to A's N/S green start. Set ϕ equal to the **free-flow travel time** on the link:

$$\phi = \frac{L}{v_f} = \frac{300}{15} = 20\ \text{s}$$

where $v_f = 15\,\text{m/s}$ (54 km/h) is the free-flow speed.

In the *uncoordinated* scenario, both intersections start their cycles at time $t = 0$ with no offset ($\phi = 0$).

1.3 Vehicle Types and Arrivals

Vehicles enter the corridor from the **south approach** of Intersection A. The combined arrival process is **Poisson** with rate $\lambda = 900\,\text{veh/h}$. Each arrival is classified by Poisson splitting:

Table 2: Vehicle type mix (southbound entry at A).

Type	**Proportion**	**Notes**
Car	85%	Standard traffic
Bus	10%	Non-preemptive queue priority (see §1.6)
Emergency	5%	Preempts current signal phase (see §1.7)

W→E cross-traffic arrives from the west as an independent Poisson process at each intersection with rate $\lambda_{\text{WE}} = 400\,\text{veh/h}$ (all regular cars, normal priority).

1.4 Turning Movements

At each intersection, a northbound vehicle may turn off the corridor. Turning decisions are made independently upon arrival at the intersection according to the following proportions:

Table 3: Turning proportions for northbound vehicles.

	Straight (N)	**Turn E**
Intersection A	0.70	0.30
Intersection B	0.80	0.20

Vehicles that turn E **leave the corridor** and do not need further modeling after exiting the intersection. Note that the demand arriving at Intersection B on the N/S corridor is *not* a fixed input—it depends on the turning proportions at A and the green time available there.

1.5 Link Model and Downstream Blocking

The link between A and B is modeled as a **point queue with finite storage capacity**. The capacity is derived from a jam density assumption:

- Link length: $L = 300\,\mathrm{m}$
- Jam density: $k_j = 133\,\mathrm{veh/km}$
- **Link capacity:** $N_{\max} = k_j \times L = 133 \times 0.3 = 40$ vehicles

Travel time on the link for a vehicle depends on the current occupancy n of the link. Use the following simple model:

$$t_{\text{travel}}(n) \;=\; \frac{L}{v_f} \;\times\; \frac{1}{1-(n/N_{\max})} \;=\; 20 \;\times\; \frac{1}{1-n/40} \quad \text{(seconds)}$$

At $n = 0$ the travel time equals the free-flow time (20 s). As n approaches $N_{\max}$, travel time increases sharply, reflecting congestion.

Downstream blocking rule. A vehicle at Intersection A may *only* proceed northward onto the link when **both** conditions are satisfied simultaneously:

1. The N/S signal at A is green, **and**
2. The current link occupancy $n < N_{\max}$.

If either condition is not met, the vehicle waits at Intersection A.

1.6 Transit Signal Priority (Buses)

Buses approaching an intersection are served before regular cars in the queue. Bus priority is **non-preemptive**: a bus moves ahead of waiting cars but does not interrupt a vehicle that is already being served (crossing the intersection).

1.7 Emergency Vehicle Preemption

When an emergency vehicle arrives at an intersection:

1. If the N/S phase is **already green**, the emergency vehicle proceeds with highest priority (ahead of all queued vehicles).
2. If the N/S phase is **not green** (i.e. W/E green or amber), the emergency vehicle **preempts** the current phase:
   - The current phase is immediately interrupted.
   - A 4 s all-red clearance interval is inserted.
   - N/S green is then activated for at least 10 s (the *emergency service time*) to allow the emergency vehicle (and any other N/S vehicles that can proceed) through.

- After the emergency vehicle clears, the signal returns to its *regular cycle*, resuming from the point in the cycle where it was interrupted.

3. Emergency vehicles are **not subject to downstream blocking** —they always proceed onto the link regardless of occupancy (but still occupy a slot on the link).

1.8 Intersection Service Time

Each vehicle requires a short time to *cross* the intersection (start-up delay + clearing the intersection). Model this as an **exponential** random variable with mean $\mu_s = 2\,\text{s}$ per vehicle. Only one vehicle per approach can cross at a time (i.e. the intersection approach acts as a single server).

2 Simulation Requirements

2.1 Random Number Streams

Use `numpy.random.default_rng()` with **separate seeds** for each random source (arrival process, vehicle classification, turning decisions, service times, link travel time noise, etc.). Document your seed assignments clearly.

2.2 Warm-up and Run Length

- Simulated duration: **2 hours** (7 200 s) per replication.
- Discard the first **15 minutes** (900 s) as warm-up.
- Number of replications: **20**.
- Report point estimates and **95% confidence intervals** for all KPIs.

3 Experiments

Run the following three scenarios:

Table 4: Experimental scenarios.

Scenario	Description	Offset ϕ	Emergency Vehicles
1	Uncoordinated, no emergencies	0 s	Disabled
2	Coordinated, no emergencies	20 s	Disabled
3	Coordinated, with emergencies	20 s	Enabled

For each scenario, collect and report the following **Key Performance Indicators (KPIs)**:

1. **Average delay per vehicle** (seconds)—separately for through-traffic (S→N) and W/E traffic.
2. **Maximum and average queue length** at each approach (N/S at A, N/S at B, W/E at A, W/E at B).
3. **Throughput**: vehicles per hour exiting northbound at B.
4. **Average link occupancy** between A and B.
5. **Downstream blocking frequency**: fraction of cycles in which at least one N/S vehicle at A was blocked due to link full.
6. **(Scenario 3 only)** Number of preemption events, and average *recovery time* (time from end of emergency service until the signal controller returns to its regular cycle point).

4 Deliverables

1. **Jupyter Notebook** (`.ipynb`)—containing all simulation code, clearly structured and documented. Your code should be runnable end-to-end.

2. **Report** (PDF, max 10 pages excluding code appendix):
   a. A brief description of your model logic and any assumptions you made beyond what is specified here.
   b. A **conceptual model diagram** (activity diagram, flowchart, or similar) showing the lifecycle of a vehicle and the signal controller logic.
   c. Results tables and/or figures for all three scenarios.
   d. A **comparative discussion**: What is the effect of coordination? How do emergency preemptions degrade corridor performance? Which KPI is most affected?
   e. A short paragraph on **verification and validation**: What sanity checks did you perform to convince yourselves the model is correct?

5 Grading Rubric

Component	**Points**
Model correctness (signal logic, preemption, priority, blocking)	35
Experimental design (seeds, warm-up, replications, CIs)	15
Code quality (readability, structure, documentation)	15
Results and analysis (tables, figures, comparative discussion)	25
Verification and validation discussion	10
Total	**100**

Parameter Summary

For easy reference, all numerical parameters are collected below:

Table 5: Complete parameter set.

Parameter	**Symbol**	**Value**
Link length	L	300 m
Free-flow speed	v_f	15 m/s (54 km/h)
Free-flow travel time	L/v_f	20 s
Jam density	k_j	133 veh/km
Link capacity	$N_{\max}$	40 vehicles
Cycle length	C	90 s
N/S green	g_{NS}	45 s
Amber (clearance)		4 s
W/E green	g_{WE}	37 s
Coordination offset	ϕ	0 s or 20 s
S entry arrival rate	λ	900 veh/h
W/E arrival rate	λ_{WE}	400 veh/h (per intersection)
Car proportion		0.85
Bus proportion		0.10
Emergency proportion		0.05
Turning straight (A)		0.70
Turn E (A)		0.30
Turning straight (B)		0.80
Turn E (B)		0.20
Intersection service	μ_s	Exp(mean = 2 s)
Emergency service time		10 s (min green)
Emergency clearance		4 s (all-red)
Warm-up		900 s
Run length		7 200 s
Replications		20

Good luck!